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RESEARCH ARTICLE

Li_2CO_3-Derived Low-Cost Li_2S for Sulfide Solid Electrolytes Exceeding 11 mS cm^{-1}

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ABSTRACT

Sulfide solid electrolytes (SSEs) hold great promise for all-solid-state batteries (ASSBs), owing to their high ionic conductivity and excellent deformability. However, their practical application is severely hindered by high cost, primarily originating from lithium sulfide (Li_2S), which accounts for ~90% of the total SSE cost. Here, we report a novel strategy to produce low-cost Li_2S from lithium carbonate (Li_2CO_3) via its reaction with ammonium thiocyanate (NH_4SCN). This reaction generates only gaseous by-products, eliminating purification procedures and enabling scalable production of high-quality Li_2S. The resulting Li_2S enables the synthesis of representative SSEs, $Li_{5.4}PS_{4.6}Cl_{0.8}Br_{0.8}$ (LPSCB) and $Li_{5.4}PS_{4.6}Cl_{1.6}$ (LPSC), with room-temperature ionic conductivities of 11.33 and 7.94 mS cm^{-1}, respectively. When coupled with $LiNbO_3$-coated $LiNi_{0.895}Co_{0.077}Mn_{0.028}O_2$ cathode, ASSBs deliver discharge capacities of 192.2 and 198.8 mAh g^{-1} at 0.1C, and retain 94.85% and 94.89% of initial capacities after 800 cycles at 1C, respectively. Cost analysis reveal that the total cost of SSEs synthesized from this Li_2CO_3-derived Li_2S is reduced by 86.6% and 88.5%, highlighting its significant techno-economic advantages for commercializing SSEs toward ASSBs.

1 | Introduction

All-solid-state batteries (ASSBs), enabled by non-flammable solid electrolytes (SEs), have emerged as promising candidates for next-generation power batteries due to their high energy density, intrinsic safety, and wide operating temperature range [1–4]. Among various SEs, sulfide SEs (SSEs, such as Li_6PS_5X, X = F, Cl, Br, I) represent a key class of inorganic electrolytes, distinguished by their exceptional ionic conductivity and favorable mechanical properties for cold-pressing processing [5–8]. However, their high production costs (>1000 USD kg^{-1})—far exceeding the commercially acceptable threshold of ~50 USD kg^{-1} [9, 10]—significantly restrict their large deployment in ASSBs. Such high material cost is mainly caused by expensive precursors lithium sulfide (Li_2S), which accounts for approximately 90% of the total cost of SSEs [11–13]. Therefore, reducing the cost of Li_2S is of paramount importance for producing cost-effective SSEs toward the commercialization of ASSBs [14].

Currently, several methods have been developed for Li_2S synthesis, including Li–S deflagration, acid–base neutralization, thermal reduction, and liquid-phase routes [15–19]. However,

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these conventional approaches suffer from substantial practical limitations. Direct deflagration of Li or LiH with sulfur is restricted by high precursor costs and severe thermal runaway risks [15]. Neutralization of LiOH with H_2S requires handling highly toxic and corrosive gases, necessitating stringent safety controls [16]. Other reported methods often face incomplete conversion, high energy consumption, expensive purification procedures, or reliance on toxic reagents [17, 18]. Consequently, developing a cost-effective, scalable, and safe route for Li_2S production is of strategic importance for sulfide-based ASSBs.

In this work, we report a novel and scalable strategy to synthesize low-cost, high-quality Li_2S from Li_2CO_3—an industry benchmark precursor (~8.56 USD kg^{-1}) via its reaction with ammonium thiocyanate (NH_4SCN)—an extremely cheap sulfur precursor (~3.56 USD kg^{-1}). This process proceeds through a stepwise metathesis reaction that generates only gaseous by-products, thereby eliminating purification procedures and enabling kilogram-scale production of Li_2S. The unpurified Li_2S exhibits a significant purity of 97%, approaching the 99.5% purity of commercial products. The simplicity and scalability of this synthesis route make it particularly attractive for industrial implementation. As-synthesized Li_2S enables the synthesis of representative SSEs, $Li_{5.4}PS_{4.6}Cl_{0.8}Br_{0.8}$ (LPSCB) and $Li_{5.4}PS_{4.6}Cl_{1.6}$ (LPSC), which exhibit high ionic conductivities of 11.33 and 7.94 mS cm^{-1}, respectively. When coupled with $LiNbO_3$-coated $LiNi_{0.895}Co_{0.077}Mn_{0.028}O_2$ (Ni90@LNO) cathode, ASSBs deliver high initial discharge capacities of 192.2 and 198.8 mAh g^{-1} at 0.1C and retain high capacity retentions of 94.85% and 94.89% of their initial capacity after 800 cycles at 1C, respectively. All-solid-state pouch cells using these SSEs also demonstrate good rate performance up to 3C. These results confirm that Li_2S produced via this low-cost route is feasible and practical for large-scale synthesis of SSEs toward practical ASSBs. Furthermore, comprehensive techno-economic analysis reveals that the total cost of SSEs synthesized using this Li_2CO_3-derived Li_2S can be reduced by up to 88.5% compared with those produced from commercially available Li_2S. This substantial cost reduction directly addresses one of the most critical barriers to the commercialization of sulfide-based ASSBs. Overall, this work stablishes a novel, scalable, and cost-effective pathway for producing Li_2S, enabling high-ionic-conductivity and low-cost SSEs for practical sulfide-based ASSBs.

2 | Results and Discussion

The Li_2S was synthesized by heating a mixture of Li_2CO_3 and NH_4SCN in a tube furnace under an Ar-flowing atmosphere, with the outlet connected to an exhaust gas treatment system, as illustrated in Figure 1a and **Experimental Section**. In this reaction, Li_2CO_3 and NH_4SCN serve as the lithium and sulfur sources, respectively. The key characteristic of this reaction is that by-products are gaseous including CO_2, NH_3, and HOCN, which eliminate the purification process of Li_2S. This overall reaction is described as Equation (1):

$$Li_2CO_3 + NH_4SCN \rightarrow Li_2S + CO_2 \uparrow + HOCN \uparrow + NH_3 \uparrow \quad (1)$$

To verify the scalability of this reaction, kilogram-scale Li_2S (denoted as Li_2S-HM) was prepared, as shown in Figure 1b. The resulting powder exhibits a uniform white appearance, indicating high quality with negligible color-imparting impurities and a low polysulfide content [20]. Notably, particle-size distribution analysis shows that Li_2S-HM exhibits finer and more uniform particles compared to commercial Li_2S (Li_2S-CM), as shown in Figure 1c,d and Table S1. This reduced particle size leads to a higher specific surface area, providing more active sites for subsequent sulfide solid electrolyte synthesis [11].

The high-resolution transmission electron microscopy (HR-TEM) image in Figure 1e reveals good crystallinity of Li_2S-HM, as evidenced by clear lattice fringes. The measured interplanar spacing of 0.333 nm corresponds well to the (111) plane of Li_2S. To further examine the microscopic morphology of the Li_2S-HM, scanning electron microscopy (SEM) was conducted (Figure 1f). The Li_2S-HM particles are uniform with well-defined edges, indicative of high crystallinity and morphological consistency, while the commercial Li_2S exhibits pronounced particle agglomeration. Energy dispersive spectroscopy (EDS) mapping further confirms the homogeneous spatial distribution of S element within in Li_2S-HM (Figure S1). Rietveld refinement of the x-ray diffraction (XRD) pattern in Figure 1h confirms that the as-synthesized powder is predominantly cubic Li_2S with an anti-fluorite structure (space group Fm-3m). To quantitatively assess the overall crystallinity of Li_2S, the degree of crystallinity (X_c) was calculated based on the area ratio of the diffraction peaks according to Equation (2) [21]:

$$X_c = \frac{I_c}{I_c + I_a} \times 100\% \quad (2)$$

Where I_c represents the integrated intensity (or area) of the crystalline peaks in the XRD pattern, and I_a represents the integrated intensity (or area) of the amorphous part. Based on this analysis, X_c value was determined to be 82.96%, with the detailed parameters summarized in Table S2. Rietveld refinement analysis suggests the lattice parameter is 5.71402 Å, in good agreement with the previously reported result of 5.71581 Å [22]. The detailed refinement data are provided in Table S3. Notably, Li_2CO_3 residual is negligible (or: below the detection limit), indicating its near-complete conversion to Li_2S. The composition and chemical states were further investigated by x-ray photoelectron spectroscopy (XPS) (Figure 1i,j). Compared with commercial Li_2S, Li_2S-HM exhibits a significantly lower concentration of polysulfide species, which is consistent with its white appearance [23].

To elucidate the reaction mechanism, the gaseous products evolved during heating were collected and analyzed using thermogravimetric-Fourier transform infrared spectroscopy (TG-IR) coupled technology. As shown in Figure 2a, four main gaseous species are identified: the absorption peak of CO_2 near 2340.2 cm^{-1}, the absorption peaks of NH_3 near 966.1 and 1721.5 cm^{-1}, the absorption peak of HOCN near 1367.2 cm^{-1}, and the absorption peak of HSCN near 1987.2 cm^{-1}. Below 200°C, the dominant gaseous by-products are NH_3, CO_2, and residual HSCN, whereas above 300°C, pronounced signals corresponding to HOCN and CO_2 are detected, indicating substitution of sulfur in SCN^- by oxygen. Notably, CO_2 evolution occurs in two distinct stages, suggesting that the conversion of Li_2CO_3 to Li_2S does not proceed in a single step.

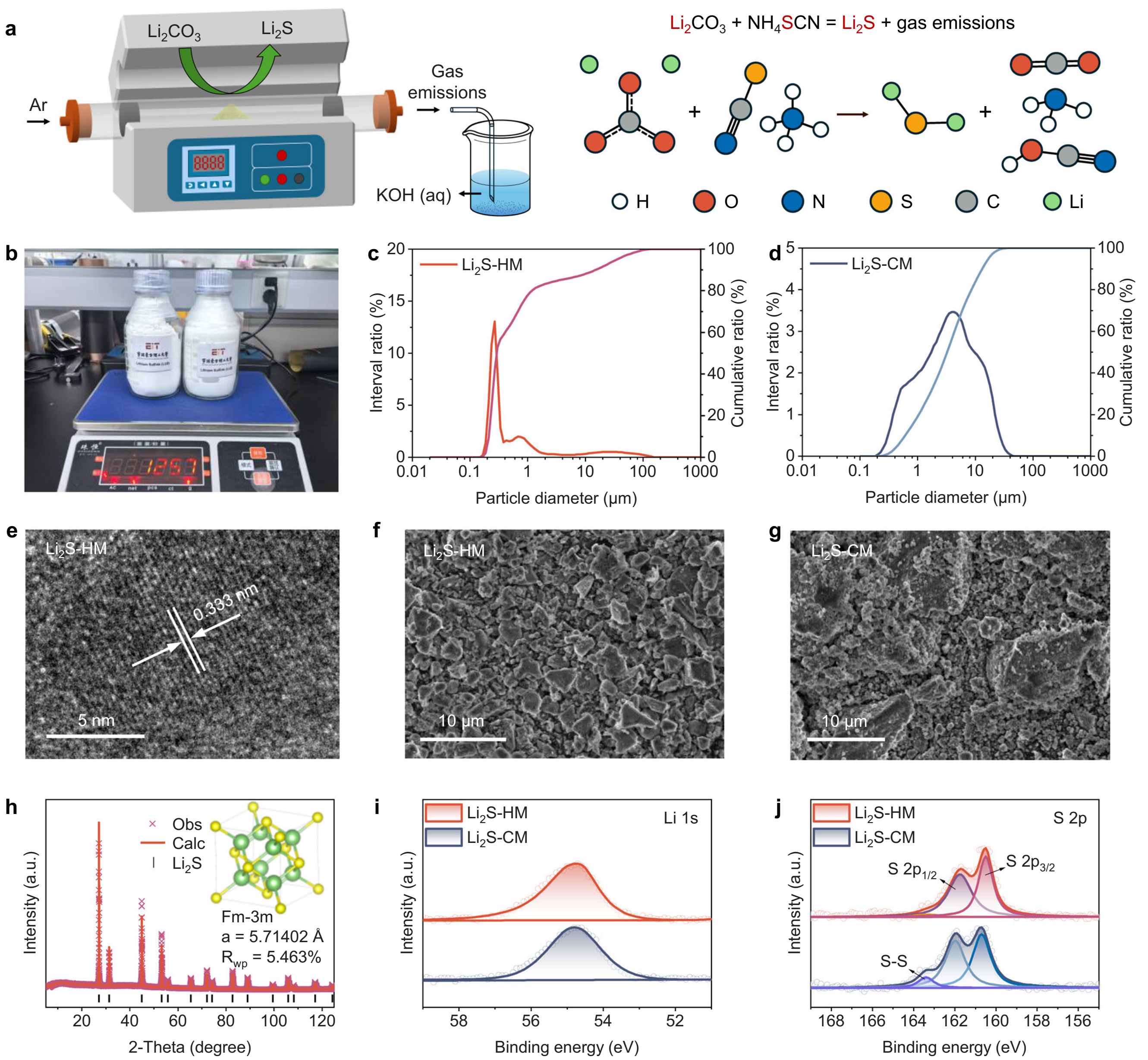

FIGURE 1 | Synthesis and Characterization of lithium sulfide (Li_2S). (a) Schematic illustration of the synthesis apparatus and reaction pathway. (b) An image showing kilogram-scale Li_2S synthesis. (c and d) Particle size distribution of home-made Li_2S (Li_2S-HM) and commercial Li_2S (Li_2S-CM). (e) HR-TEM image of Li_2S-HM. (f and g) SEM images of Li_2S-HM (f) and Li_2S-CM (g). (h) Rietveld refinement of XRD pattern. (i and j) XPS spectra of Li_2S-HM in comparison with Li_2S-CM: (i) Li 1s and (j) S 2p.

Based on these observations, a stepwise reaction pathway is proposed (Figure 2b). In the presence of excess of NH_4SCN, Li_2CO_3 initially undergoes partial metathesis to form LiSCN and $(NH_4)_2CO_3$ (Equation 3). Upon heating below 200 °C, $(NH_4)_2CO_3$ decomposes into NH_3, CO_2, and H_2O, accounting for the first stage of CO_2 release (Equation 4). At higher temperatures (>300 °C), CO_3^{2-} from remaining Li_2CO_3 nucleophilically attacks the SCN^- in LiSCN, producing Li_2S, HOCN, and CO_2, which corresponds to the second stage of CO_2 release (Equation 5). The generated HOCN reacts with NH_3 to form ammonium cyanate (NH_4OCN), which then isomerizes into urea ($OC(NH_2)_2$) via the “Wöhler Synthesis” (Equation 6) [24]. Nonetheless, eco-harmful emissions remain a concern from an environmental perspective [25]. The formation of urea is confirmed by liquid chromatography-mass spectrometry (LC-MS) analysis of the collected by-products, as provided in Figures S2 and S3. The stepwise reactions are described as below:

$$NH_4SCN + 0.5Li_2CO_3 \rightarrow 0.5(NH_4)_2CO_3 + LiSCN \tag{3}$$

$$0.5(NH_4)_2CO_3 \rightarrow NH_3\uparrow + 0.5H_2O + 0.5CO_2\uparrow \tag{4}$$

$$LiSCN + 0.5H_2O + 0.5Li_2CO_3 \rightarrow Li_2S + HOCN\uparrow + 0.5CO_2\uparrow \tag{5}$$

$$HOCN + NH_3 \rightarrow NH_4OCN \rightarrow OC(NH_2)_2\,(\text{Urea}) \tag{6}$$

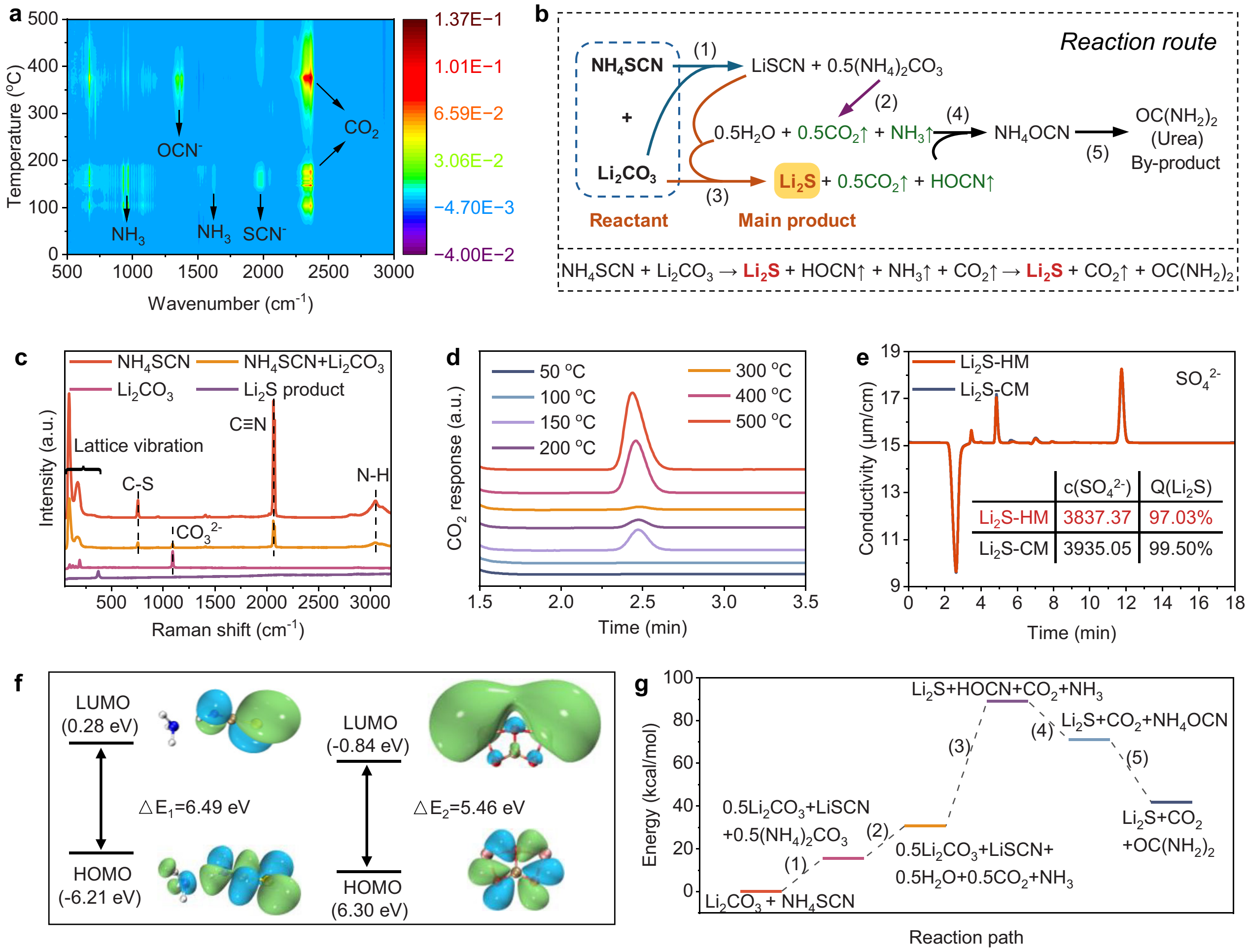

FIGURE 2 | Investigation of reaction pathway and mechanism. (a) TG-IR spectra of Li_2CO_3 and NH_4SCN with a mole ratio of 1:1. (b) Reaction route from Li_2CO_3 and NH_4SCN to Li_2S. (c) Raman spectra of NH_4SCN, Li_2CO_3, $NH_4SCN + Li_2CO_3$, and Li_2S product. (d) GC-FID-TCD spectra of Li_2CO_3 and NH_4SCN with a mole ratio of 1:1. (e) IC spectra of Li_2S-HM and Li_2S-CM. (f) HOMO−LUMO of NH_4SCN and Li_2CO_3 molecules. (g) Relative free energy diagram during the reaction path.

Raman spectroscopy was employed to examine NH_4SCN, Li_2CO_3, a ball-milled mixture of Li_2CO_3 and NH_4SCN in a 1:1 molar ratio, and the final Li_2S product (Figure 2c). NH_4SCN exhibits characteristic C—S stretching (~750 cm^{-1}), N—H stretching (~3200 cm^{-1}), and lattice vibration modes below 300 cm^{-1}, while Li_2CO_3 shows a distinct C—O stretching mode at ~1167 cm^{-1}. These features coexist in the ball-milled precursor mixture. In contrast, the final product displays only a Li—S stretching vibration at ~386 cm^{-1}, with no residual signals from the reactants, confirming complete conversion. Gas chromatography equipped with flame ionization and thermal conductivity detectors (GC–FID–TCD) further verified CO_2 evolution during heating from 50°C to 400°C (Figure 2d). Two distinct CO_2 release peaks at approximately 150°C and 400°C are observed, consistent with the TG-IR results in Figure 2a.

To quantify the product purity, 100 mg each of commercially available Li_2S (99.5% purity) and the Li_2S obtained in this work were dissolved separately in an excess of 3 wt% H_2O_2 solution for complete oxidation, described as Equation (7) [26]:

$$Li_2S + 4H_2O_2 \rightarrow Li_2SO_4 + 4H_2O \qquad (7)$$

The resulting SO_4^{2-} concentration was determined by ion chromatography (IC). Li_2S-HM exhibits a purity of ~97.03% (Figure 2e and Table S4), lower than the 99.5% purity of commercial Li_2S but fully sufficient for preparing high-performance SSEs. To investigate the composition of impurities, Raman spectroscopy was performed on the Li_2S product. The spectrum revealed weak peaks at 1545, 1920, 2053, 2261, and 2384 cm^{-1}, corresponding to the stretching vibrations of C═C, —N═C═O, C≡C, —C≡N, and O═C═O, respectively, as shown in Figure S4. Furthermore, the magnified XRD pattern in Figure S5 exhibited no detectable crystalline diffraction peaks, indicating that these impurities exist primarily as amorphous C—N—O compounds.

To further understand the reaction pathway, frontier molecular orbital (FMO) analysis was conducted using density functional theory (DFT) [27]. As shown in Figure 2f, NH_4SCN and Li_2CO_3 exhibit large HOMO−LUMO gaps of 6.49 and 5.46 eV, respectively. This suggests that the reaction does not proceed directly through orbital overlap and electron transfer but is more likely to occur via ion exchange [28]. Figure 2g shows the relative free energy diagram of the multi-step reaction pathway, demonstrating this reaction proceeds through a series of low-energy intermediates, with a relatively small free energy change, thereby

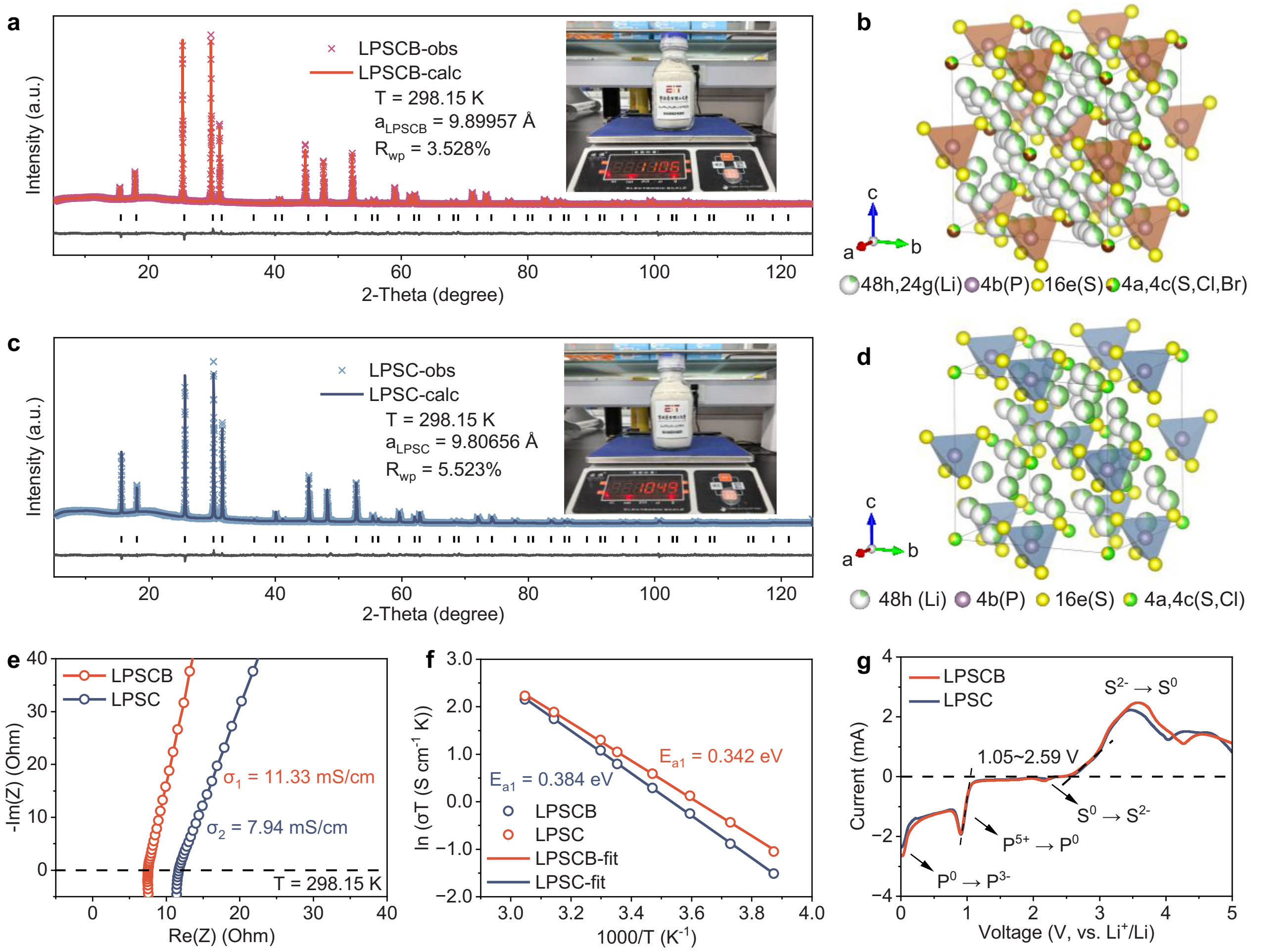

FIGURE 3 | Synthesis and characterization of sulfide electrolytes. (a–d) Rietveld refinement and crystal structure model of XRD patterns of (a and b) LPSCB and (c and d) LPSC. (e) Nyquist plots, (f) corresponding Arrhenius plots, and (g) LSV plots of LPSCB and LPSC.

providing a kinetically feasible pathway for the reaction [29]. This also explains why the decomposition temperature of Li_2CO_3 is reduced from 899.9°C to below 500°C [30]. Overall, this reaction pathway enables a low-cost and scalable synthesis of high-quality Li_2S, providing a solid foundation for the production of economically viable SSEs for ASSBs.

To evaluate the practical applicability of as-synthesized Li_2S, two representative halogen-rich SSEs, $Li_{5.4}PS_{4.6}Cl_{0.8}Br_{0.8}$ (LPSCB) and $Li_{5.4}PS_{4.6}Cl_{1.6}$ (LPSC), were prepared on a kilogram scale. Since both SSEs are free of costly metallic elements (Ge, Ta etc.), they hold a compelling cost potential for future applications [31]. XRD analysis confirmed the high crystallinity of both SSEs. The Rietveld refinement results and corresponding crystal structure of LPSCB and LPSC are presented in Figure 3a–d, with corresponding refinement parameters summarized in Tables S5 and S6. Le Bail fitting indicates both patterns belong to the F-43m space group [32]. The excellent agreement between experimental and calculated patterns, along with the absence of impurity peaks, demonstrates the high phase purity of as-prepared LPSCB and LPSC.

Electrochemical impedance spectroscopy (EIS) was employed to measure the ionic conductivity of the SSEs (Figure 3e). At 25°C, LPSCB and LPSC exhibit ionic conductivities of 11.33 and 7.94 mS cm^{-1}, respectively. The Br dopants with larger ionic radius reduces the mechanical stability of the sulfide tetrahedral ion clusters and enhances the paddle-wheel effect, consequently leading to the higher ionic conductivity observed in LPSCB compared to LPSC [33]. From the temperature-dependent conductivity fitted by the Arrhenius equation in Figure 3f, the activation energies for LPSCB and LPSC are determined to be 0.342 and 0.384 eV, respectively, indicating faster Li^+ migration in the mixed-halide SSEs. Direct current (DC) polarization curves at 0.1–0.5 V suggests that the electronic conductivities remained at a very low level of approximately 1.5×10^{-8} S cm^{-1} and 3.2×10^{-8} S cm^{-1} for LPSCB and LPSC (Figure S6), respectively. This result further confirms that the synthesized Li_2S does not contain residual electronically conductive impurities [20, 34].

Furthermore, linear sweep voltammetry (LSV) was conducted to determine the electrochemical stability window of SSEs, as shown in Figure 3g. Both curves showed similar trends, with an electrochemical stability window ranging from 1.05 to 2.59 V. The LSV profiles of LPSCB and LPSC nearly overlapped within the 0.5–3.5 V range. Above 3.5 V, LPSC exhibits an oxidation peak at a lower potential and a smaller integrated area compared to LPSCB, suggesting that LPSC possesses better oxidation stability. When scanning from the open-circuit voltage (2.37 V) toward lower potentials, the reduction peak corresponding to the conversion

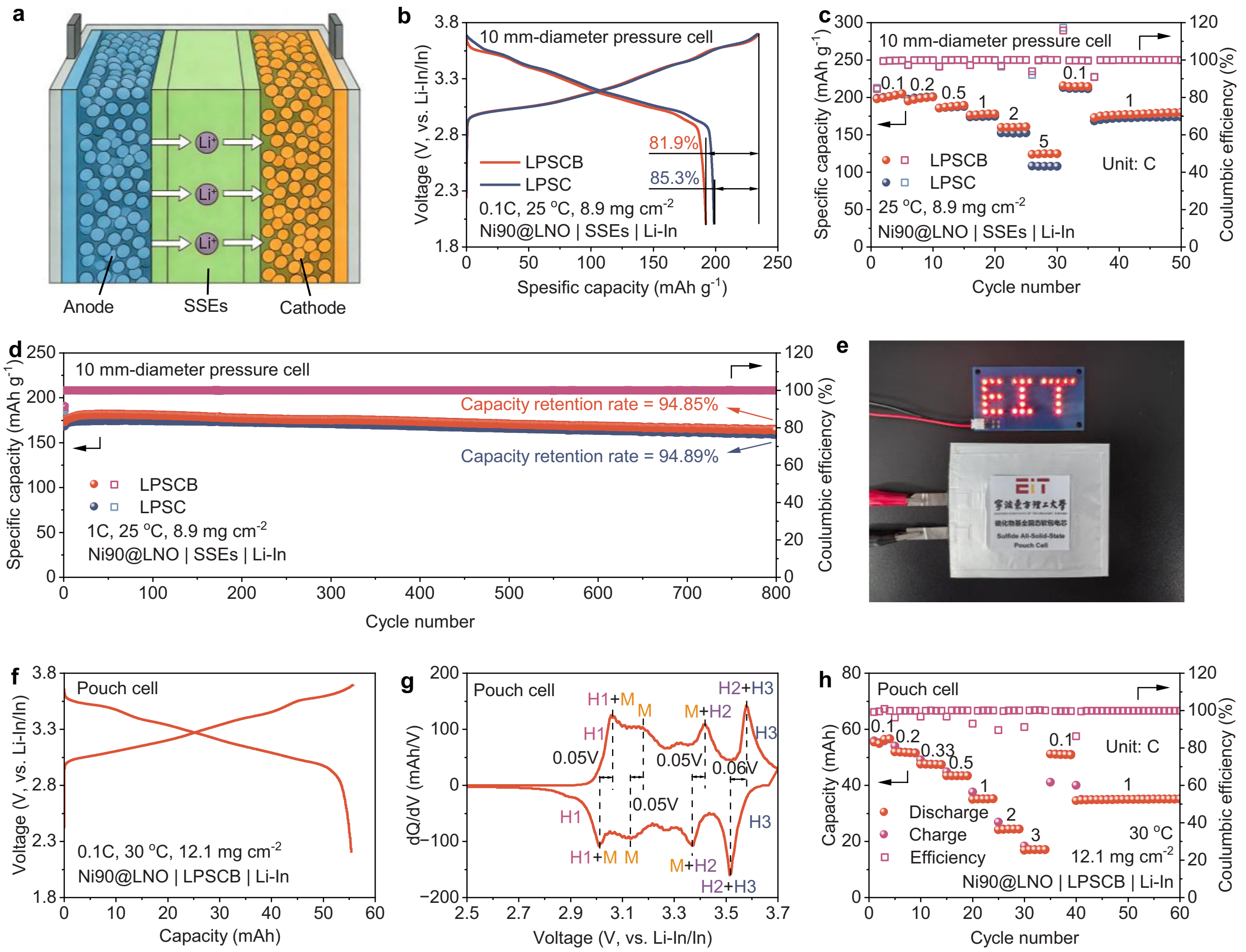

FIGURE 4 | SSE-based ASSBs and their electrochemical performance. (a) Schematic diagram of the Ni90@LNO | SSEs | Li-In cell configuration. (b) Galvanostatic charge-discharge curves at 0.1C, (c) rate performance, (d) long-term cycling performance at 1C for 10 mm-diameter pressure cells employing LPSCB and LPSC. (e) Photograph of a pouch cell powering LED bulbs. (f) Charge–discharge curves at 0.1C, (g) dQ/dV plot, and (h) rate performance of the pouch cell with a Ni90@LNO | LPSCB | Li-In configuration after formation at 30°C.

of residual S^0 to S^{2-} occurs earlier for LPSC.22 V) than for LPSCB (2.16 V). At approximately 1.05 V, both electrolytes undergo nearly simultaneous reduction of P^{5+} to P^0, followed by further reduction from P^0 to P^{3-}. Scanning from the open-circuit voltage toward higher potentials mainly involves the oxidation of S^{2-} to S^0 [35]. As previously reported, despite the narrow thermodynamic stability window of thiophosphate-based solid electrolytes, stable cycling of ASSBs can still be achieved beyond this range. This is attributed to kinetic stabilization enabled by the formation of a passivation interphase (SEI/CEI) at the electrode/electrolyte interfaces, which effectively suppresses further decomposition while maintaining ionic conduction [36]. Raman spectroscopy was further employed to investigate the local structure of as-synthesized SSEs (Figure S7). The characteristic stretching vibrations of $[PS_4]^{3-}$ units appears at 424.8 cm^{-1} for LPSCB and 428.9 cm^{-1} for LPSC. The redshift observed for LPSCB reflects lattice expansion induced by partial Br substitution, which is consistent with its reduced activation energy and enhanced ionic conductivity [8]. Overall, these results demonstrate that the as-synthesized Li_2S enables the scalable fabrication of high-performance sulfide SSEs with room-temperature ionic conductivities exceeding 11 mS cm^{-1}, validating its strong potential for practical solid-state battery applications.

To evaluate the electrochemical performance of the as-synthesized SSEs, ASSBs with a $LiNbO_3$-coated $LiNi_{0.895}Co_{0.077}Mn_{0.028}O_2$ (Ni90@LNO) cathode and a Li–In alloy anode in 10 mm-diameter pressure cells were assembled (Figures 4a and S8). Figure 4b presents the initial charge-discharge profiles of the ASSBs at a rate of 0.1C. ASSBs with LPSCB and LPSC deliver initial capacities of 192.2 mAh g^{-1} (1.714 mAh cm^{-2}) and 198.8 mAh g^{-1} (1.773 mAh cm^{-2}), with corresponding initial Coulombic efficiencies of 81.9% for LPSCB and 85.2% for LPSC, respectively. To verify the reproducibility of the cells' performance, five groups of ASSBs were fabricated with error bars included (Figure S9). The results show that the standard deviations of the capacities were all within 2 mAh g^{-1}. The slightly lower initial Coulombic efficiency of LPSCB is mainly attributed to the easier oxidation of Br^- compared to Cl^-, which leads to the formation of a thicker cathode-electrolyte interphase (CEI), thus consuming more active lithium [37].

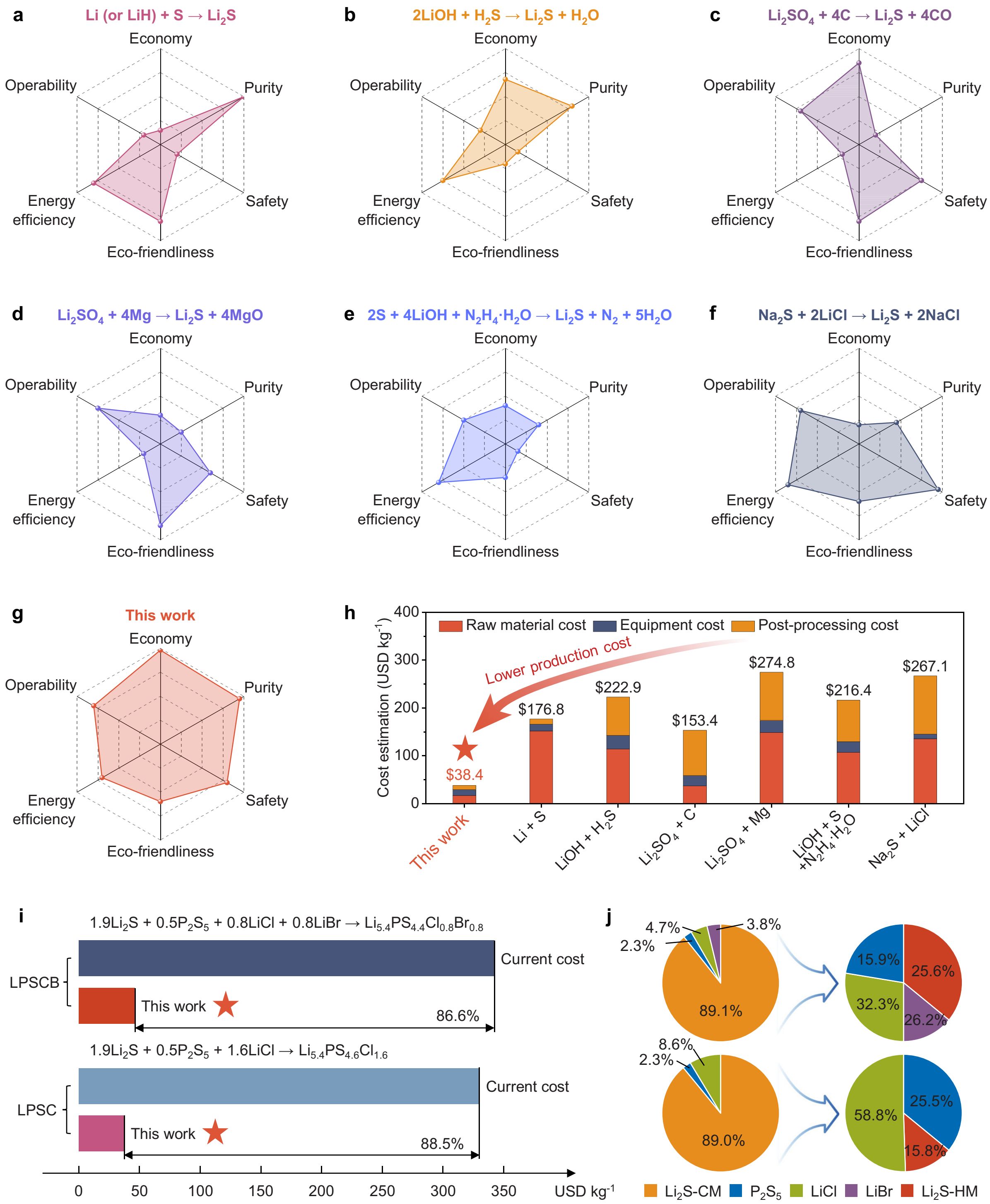

FIGURE 5 | Feasibility and techno-economic analysis. (a–g) Radar chart of six metrics for different Li_2S synthesis methods. (h) Cost estimation of Li_2S in different synthesis methods. (i) Cost evolution of SSEs relative to comparable commercial products. (j) Evolution of Li_2S cost contribution in the LPSCB and LPSC SSEs.

At low current densities below 1C, ASSBs with both SSEs deliver comparable capacities; however, at rates above 1C, LPSCB-based ASSBs exhibit significantly higher capacity than LPSC counterparts (Figure 4c), which is ascribed to the higher ionic conductivity of LPSCB. Even at 5C, ASSBs with LPSCB and LPSC still deliver high reversible capacities of 123.9 and 108.3 mAh g^{-1}, respectively. Figure 4d displays the long-term cycling performance at 1C. After 800 cycles, ASSBs employing LPSCB and LPSC retain 94.85% and 94.89% of their initial capacity, respectively. These excellent electrochemical properties demonstrate the stable practical applicability of SSEs synthesized from Li_2S.

To demonstrate practical applicability, an all-solid-state pouch cell with a structure of Ni90@LNO | LPSCB | Li-In was fabricated and successfully powered a red LED (Figure 4e). After formation at 30°C, the all-solid-state pouch cells present a charge–discharge profile with a reversible capacity of 55.4 mAh (2.2 mAh cm^{-2}) (Figure 4f). The dQ/dV analysis in Figure 4g shows well-defined peaks, corresponding to the characteristic "H1 → M → H2 → H3" phase transitions. The high coulombic efficiency of 99.3% and the extremely low polarization of ~0.05 V (close to ~0.02 V of the pressure cells in Figure S10) attest to the high electrochemical reversibility and superior electrode kinetics in all-solid-state pouch cells. To further elucidate the kinetic performance of the pouch cell, its rate capability is presented in Figure 4h. Notably, the cell maintains 30.32% of its 0.1C capacity (55.4 mAh) even at a high rate of 3C (16.8 mAh), with a retention of 63.36% at 1C (35.1 mAh). The superior performance of the pouch cell further confirms the practical utility of the Li_2S-derived SSEs.

From industrialization and engineering perspective, we establish a six-dimensional evaluation framework encompassing economy, purity, safety, environmental friendliness, energy efficiency, and operability to systematically assess mainstream Li_2S production routes. Based on their industrial properties, different synthesis routes were objectively evaluated using a 100-point scale, and the results are presented as radar charts (Figure 5a–g) to visualize the advantages, disadvantages, and overall profile of each method. The Li-S direct deflagration method (Figure 5a), although adopted in early industrial practice, is intrinsically limited by high precursor costs and severe safety risks arising from intense heat release and uncontrollable kinetics. This method is suitable for cost-insensitive fields with high-quality requirements [38]. The LiOH-H_2S neutralization route (Figure 5b) benefits from low material cost but is hampered by the toxicity, corrosion, and environmental hazards of H_2S, as well as concerns over product purity and process stability. Carbothermal reduction (Figure 5c) is economically attractive in terms of raw materials; however, insufficient product purity and the need for costly secondary purification significantly pose significant challenges. The magnesium-thermal route (Figure 5d) suffers from high energy consumption, low yield, and expensive reagents, while the hydrazine hydrate reduction method (Figure 5e) involves complex processing, solvent recovery challenges, moisture-sensitive reagents, and severe equipment corrosion. Conventional metathesis routes using $Na_2S{\cdot}xH_2O$ (Figure 5f) are also economically unfavorable due to the high cost of sulfide precursors.

In contrast, the strategy developed in this work employs low-cost Li_2CO_3, derived from salt-lake brines [39] or spodumene [40], along with industrially abundant NH_4SCN (Figure 5g). The stepwise metathesis mechanism effectively lowers the reaction activation barrier, offering clear advantages in raw-material cost, safety, and scalability. The high-temperature sintering process may result in high energy consumption, and emissions of gases such as NH_3, HOCN, and CO_2 may pose additional challenges. Although the high-temperature and exhaust gas treatment present certain difficulties, this route demonstrates balanced and competitive performance across all six evaluation dimensions.

To quantitatively compare production costs, we estimated the expenses for manufacturing 100 tons of Li_2S by dividing costs into raw materials, equipment, and post-treatment (Figure 5h and Table S7). While the Li-S deflagration method incurs minimal post-treatment costs, it is dominated by expensive raw materials. Carbothermal reduction suffers from high purification costs, and other routes—including acid–base neutralization, magnesium-thermal reduction, hydrazine reduction, and conventional metathesis—require substantial investment in both raw materials and post-treatment. By contrast, this Li_2CO_3-derived Li_2S via the stepwise metathesis reaction shows no evident cost bottleneck in any category, enabling a projected total cost below 50 USD kg^{-1}, consistent with commercial targets [14].

Based on laboratory-scale prices (Table S8), the bulk prices of raw materials required for synthesizing LPSCB and LPSC electrolytes (Figure 5i) were estimated using the general logarithmic correlation between unit price and purchase quantity of chemicals, as expressed in Equation (8) [41]:

$$\mathrm{Log}_{10}\mathrm{P} = \mathrm{Log}_{10}\mathrm{a} + \mathrm{b} \times \mathrm{Log}_{10}Q \tag{8}$$

where P and Q represent the unit price and purchase quantity of chemical reagents, respectively, while a and b are constants for a given raw material. The results show that using Li_2CO_3-derived Li_2S as demonstrated in this work, the cost of LPSCB and LPSC electrolytes decreases from 345 and 328 to 46 and 39 USD kg^{-1}, corresponding to the cost reductions of 86.6% and 88.5%, respectively. Notably, the cost contribution of Li_2S drops from ~89% to 25.6% for LPSCB and 15.8% for LPSC (Figure 5j), indicating that Li_2S is no longer the dominant cost factor and enabling realistic large-scale commercialization of SSEs for ASSBs.

3 | Conclusion

In summary, we have developed a scalable and cost-effective strategy for synthesizing high-quality Li_2S using industry-benchmark Li_2CO_3 and ammonium thiocyanate (NH_4SCN) as low-cost precursors. The gas-releasing stepwise metathesis mechanism eliminates purification process, lowers reaction barriers, and enables kilogram-scale synthesis of high-quality Li_2S. The generation of eco-harmful gases could yet present new challenges for large-scale expansion. The as-synthesized Li_2S enable the successful synthesis of SSEs—$Li_{5.4}PS_{4.6}Cl_{0.8}Br_{0.8}$ and $Li_{5.4}PS_{4.6}Cl_{1.6}$—which exhibit high ionic conductivities of 11.33 and 7.94 mS cm^{-1}, respectively. ASSBs incorporating these SSEs demonstrate excellent long-term cycling stability and superior rate capability, which are further validated in practical all-solid-state pouch-cell configurations. Finally, a techno-economic analysis shows up to ~88% cost reduction in total SSE cost, reaching commercial targets. This scalable synthetic strategy not only significantly reduces the production cost of Li_2S but also facilitates the industrialization of SSEs toward their ASSBs.

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Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Mengfei Zhu: writing–original draft, data curation, visualization. **Shengjie Xia**: validation, data curation. **Chao Wang**: writing–original draft, visualization. **Haoxiong Hu**: validation, investigation. **Ziqing Wang**: validation, visualization. **Mingfeng Wei**: visualization. **Tingting Liu**: visualization. **Suzhe Liang**: visualization. **Guantai Hu**: validation. **Luting Xie**: validation, visualization. **Kaiyong Tuo**: validation, visualization. **Zhimin Zhou**: validation, visualization. **Shutao Zhang**: validation. **Jian Hong**: validation. **Xueliang Sun**: writing–review and editing, resources, project administration, funding acquisition, conceptualization. **Changhong Wang**: conceptualization, funding acquisition, writing–review and editing, project administration, resources.

Data Availability Statement

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supporting Information.

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Supporting Information

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Supporting File 1: anie72627-sup-0001-SuppMat.docx.